MARKING SCHEME

Model Paper — 04

Third Term Test (Full Syllabus)

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Note (for teachers):

- Part A: 2 marks for each question (generally 1 mark for the correct method and 1 mark for the correct final answer). — Total 40 marks
- Part B: 10 marks for each question. — Total 60 marks. The marks for each step are shown in the right-hand column.
- Award full marks for correct alternative methods. Accept a reasonable range for values read from graphs.

Part A — Answers

1. $\sqrt{3249}$: $32 \rightarrow 5^2 = 25$, remainder 7 ; $749 \rightarrow 107 \times 7 = 749$, remainder 0

Final answer: 57

(2 marks)

2. $\frac{5}{3} \times \frac{6}{5} = 2$; $\frac{15}{4} - 2 = \frac{7}{4}$

Final answer: $\frac{7}{4} = 1\frac{3}{4}$

(2 marks)

3. $(3x)^2 + 2(3x)(2y) + (2y)^2$

Final answer: $9x^2 + 12xy + 4y^2$

(2 marks)

4. $2x^2 + 10x - 3x - 15 = 2x(x + 5) - 3(x + 5)$

Final answer: $(2x - 3)(x + 5)$

(2 marks)

5. L.C.M. of the coefficients = 24 ; x^3, y^2

Final answer: $24x^3y^2$

(2 marks)

6. $\frac{a}{a-b} - \frac{b}{a-b} = \frac{a-b}{a-b}$

Final answer: 1

(2 marks)

7. $\frac{\theta}{360} \times \frac{22}{7} \times 7 \times 7 = 38.5 \Rightarrow \frac{\theta}{360} \times 154 = 38.5 \Rightarrow \frac{\theta}{360} = \frac{1}{4}$

Final answer: 90°

(2 marks)

8. Exterior angle = $\hat{A} + \hat{B} \Rightarrow 3x = 117 \Rightarrow x = 39$; $\hat{B} = 2 \times 39$

Final answer: 78°

(2 marks)

9. $\log_5 5^3 - \log_2 2^{-3} = 3 - (-3)$

Final answer: 6

(2 marks)

10. $\lg 4 + \lg 5^2 = \lg (4 \times 25) = \lg 100$

Final answer: 2

(2 marks)

11. $a + 4d = 17$; $a + 9d = 32 \Rightarrow 5d = 15$

Final answer: d = 3

(2 marks)

12. $A = \{1, 2, 3, 4, 6, 12\} \Rightarrow n(A) = 6$; $n(A') = 12 - 6$

Final answer: 6

(2 marks)

13. Loss = $4\,800 \times \frac{15}{100} = 720$; $4\,800 - 720$

Final answer: Rs. 4 080

(2 marks)

14. Pump-hours = $8 \times 9 = 72$; $72 \div 6$

Final answer: 12 hours

(2 marks)

15. $r^2 = \frac{A}{\pi}$

Final answer: $r = \sqrt{\frac{A}{\pi}}$

(2 marks)

16. $3x - 3 > 2x + 4 \Rightarrow x > 7$

Final answer: 8

(2 marks)

17. $27 \times \frac{360}{54}$

Final answer: 180

(2 marks)

18. Favourable outcomes {5, 6} ; $\frac{2}{6}$

Final answer: $\frac{1}{3}$

(2 marks)

19. Area = base × perpendicular height ⇒ 48 = 8 × h

Final answer: 6 cm

(2 marks)

20. 360 = 9 × 8 × h ⇒ 72h = 360

Final answer: 5 cm

(2 marks)

Part B — Answers and Mark Allocation

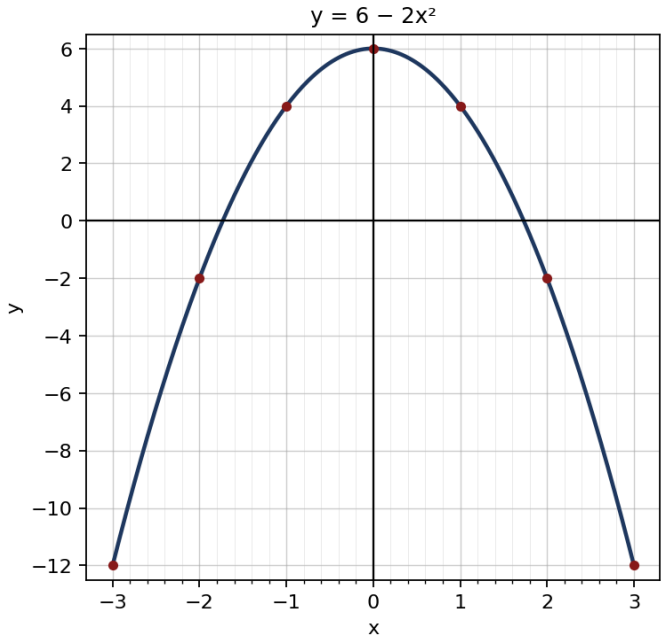
Question 1

(a)	
① × 2: 4x − 2y = 14	**1**
Adding with ②: 7x = 35	**1**
x = 5	**1**
By substitution: 10 − y = 7 ⇒ y = 3	**1**
(b)	
(x + 6)(x − 4) = 0	**1**
x = 4 or x = −6	**1**
(c)	
(i) $x + x^2 = 42 \Rightarrow x^2 + x - 42 = 0$	**1**
(ii) (x + 7)(x − 6) = 0	**1**
x = 6 or x = −7	**1**
(iii) x = 6 ⇒ 2(6) − 1 = 11	**1**

Total marks: 10

Question 2

(a)	
$x = -1 \Rightarrow y = 4$; $x = 3 \Rightarrow y = -12$	**1**
(b)	
Correct scale and axes	**1**
Plotting the 7 points correctly	**1**
Smooth curve	**1**



(c)	
(i) Maximum value 6 ; turning point (0, 6)	**1**
(ii) Reading the points where the curve cuts the x-axis	**1**
$x \approx -1.7$ or $x \approx 1.7$ (accept ±1.7 – ±1.8)	**1**
(iii) $-1.7 \leq x \leq 1.7$	**1**
(iv) $y \approx -6.5$ (accept −6.3 – −6.7)	**1**
(d)	
$y = -2x^2$	**1**

Total marks: 10

Question 3

(a)	
Volume = $\pi(R^2 - r^2)h = \frac{22}{7} \times (49 - 12.25) \times 20$	**1**
$= \frac{22}{7} \times 36.75 \times 20$	**1**
$= 2\ 310\ cm^3$	**1**
(b)	
$2\pi Rh = 2 \times \frac{22}{7} \times 7 \times 20$	**1**
$= 880\ cm^2$	**1**
(c)	
$2\ 310 \times 8 = 18\ 480\ g$	**1**
$= 18.48\ kg$	**1**
(d)	
Volume of water = $\frac{22}{7} \times 3.5 \times 3.5 \times 20 = 770\ cm^3$	**1**
= 0.77 litres	**1**
$7.7 \div 0.77 = 10$ pipes	**1**

Total marks: 10

Question 4

(a)	
Figure showing A, B, C, E, F, O and the right angles correctly	**1**
(b) In triangles BFC and CEB,	
$B\hat{F}C = C\hat{E}B = 90°$; BC is the common side	**1**
$F\hat{B}C = E\hat{C}B$ (base angles of the isosceles triangle, since AB = AC)	**1**
$\therefore \Delta BFC \equiv \Delta CEB$ (A.A.S.)	**1**
(c)	
Corresponding sides of congruent triangles $\Rightarrow$ CF = BE	**1**
(d)	
Corresponding angles $\Rightarrow F\hat{C}B = E\hat{B}C$, that is $O\hat{C}B = O\hat{B}C$	**1**
Triangle OBC is isosceles $\Rightarrow$ OB = OC	**1**
(e)	
$A\hat{B}C = (180° - 50°) \div 2 = 65°$; in triangle ABE, $A\hat{B}E = 90° - 50° = 40°$	**1**
$O\hat{B}C = 65° - 40° = 25° = O\hat{C}B$	**1**
$B\hat{O}C = 180° - 50° = 130°$	**1**

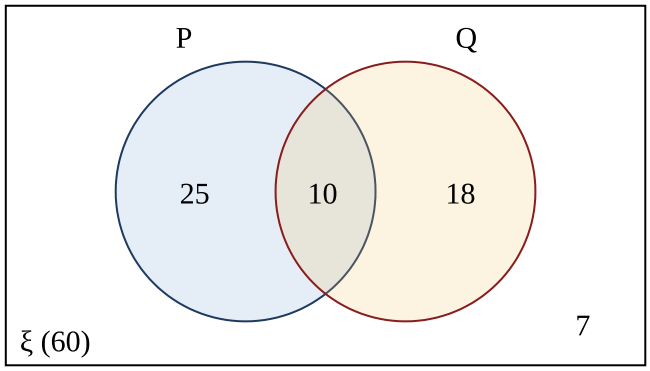
Total marks: 10

Question 5

(a)	
20 – 30	**1**
(b)	
Class mid-values: 5, 15, 25, 35, 45	**1**
fx: 30, 135, 375, 420, 360	**1**
Σfx = 1 320	**1**
Mean = 1 320 ÷ 50 = 26.4 minutes	**1**
(c)	
Less than 20 minutes = 6 + 9 = 15	**1**
$\frac{15}{50} \times 100 = 30\%$	**1**
(d)	
Total decrease in time = 8 × 10 = 80	**1**
New total = 1 320 − 80 = 1 240	**1**
New mean = 1 240 ÷ 50 = 24.8 minutes	**1**

Total marks: 10

Question 6

(a)	
	
ξ and the two intersecting circles	**1**
10 in the intersection ; P only 25, Q only 18	**1**
7 outside	**1**
(b)	
$n(P \cup Q) = 35 + 28 - 10$	**1**
$= 53$	**1**
(c)	
$60 - 53 = 7$	**1**
(d)	
(i) Only one = 25 + 18 = 43	**1**
$P = \frac{43}{60}$	**1**
(ii) Q only = 28 − 10 = 18	**1**
$P = \frac{18}{60} = \frac{3}{10}$	**1**

Total marks: 10